

ModelCardSec Security Model Card

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model	aggregate_risk	risk_band	attack_success_proxy
Logistic Regression	16.34	Low	0.0343
Random Forest	20.21	Low	0.1201
Decision Tree	18.38	Low	0.0585

Logistic Regression: aggregate risk 16.34 (Low)

Robustness - risk 3.16, status ok

Finding: Baseline accuracy: 0.977.

Finding: Noisy-input accuracy: 0.965.

Finding: Prediction flip rate under perturbation: 0.012.

Recommendation: Keep robustness checks in continuous evaluation as the data distribution changes.

Membership - risk 3.43, status ok

Finding: Confidence-based membership AUC: 0.483.

Finding: Membership advantage proxy: 0.034.

Finding: Train-test mean confidence gap: -0.002.

Recommendation: Continue tracking confidence gaps for each model release.

Calibration - risk 4.70, status ok

Finding: Expected calibration error: 0.019.

Finding: Brier score: 0.017.

Recommendation: Keep calibration in release gates, especially after retraining or data drift.

Pii - risk 30.00, status ok

Finding: PII-like patterns in sampled training text: {'email': 1, 'phone': 1, 'ssn': 0, 'credit_card_like': 0, 'ip_address': 0}.

Finding: PII-like patterns in sampled outputs: {'email': 1, 'phone': 0, 'ssn': 0, 'credit_card_like': 0, 'ip_address': 0}.

Finding: Exact training-output repeats: 0.

Recommendation: Maintain PII scans in data ingestion and output monitoring pipelines.

Unsafe Outputs - risk 33.33, status ok

Finding: Unsafe-output category hits: {'credential_exposure': 1, 'malware_or_intrusion': 0, 'self_harm': 0, 'unsafe_tool_use': 0}.

Finding: Example hits: credential_exposure: Potential secret: api_key=do-not-ship-this-example

Recommendation: Continue red-team prompt evaluation for every prompt/template or model change.

Drift - risk 34.54, status ok

Finding: Maximum feature PSI: 0.173.

Finding: Mean feature PSI: 0.072.

Finding: Top drifted features: symmetry error=0.173, smoothness error=0.168, fractal dimension error=0.145, worst smoothness=0.141, texture error=0.108

Recommendation: Review moderate drift and evaluate model performance on the shifted slice.

Random Forest: aggregate risk 20.21 (Low)

Robustness - risk 7.13, status ok

Finding: Baseline accuracy: 0.971.

Finding: Noisy-input accuracy: 0.947.

Finding: Prediction flip rate under perturbation: 0.035.

Recommendation: Keep robustness checks in continuous evaluation as the data distribution changes.

Membership - risk 12.01, status ok

Finding: Confidence-based membership AUC: 0.560.

Finding: Membership advantage proxy: 0.120.

Finding: Train-test mean confidence gap: 0.025.

Recommendation: Continue tracking confidence gaps for each model release.

Calibration - risk 13.81, status ok

Finding: Expected calibration error: 0.055.

Finding: Brier score: 0.026.

Recommendation: Keep calibration in release gates, especially after retraining or data drift.

Pii - risk 30.00, status ok

Finding: PII-like patterns in sampled training text: {'email': 1, 'phone': 1, 'ssn': 0, 'credit_card_like': 0, 'ip_address': 0}.

Finding: PII-like patterns in sampled outputs: {'email': 1, 'phone': 0, 'ssn': 0, 'credit_card_like': 0, 'ip_address': 0}.

Finding: Exact training-output repeats: 0.

Recommendation: Maintain PII scans in data ingestion and output monitoring pipelines.

Unsafe Outputs - risk 33.33, status ok

Finding: Unsafe-output category hits: {'credential_exposure': 1, 'malware_or_intrusion': 0, 'self_harm': 0, 'unsafe_tool_use': 0}.

Finding: Example hits: credential_exposure: Potential secret: api_key=do-not-ship-this-example

Recommendation: Continue red-team prompt evaluation for every prompt/template or model change.

Drift - risk 34.54, status ok

Finding: Maximum feature PSI: 0.173.

Finding: Mean feature PSI: 0.072.

Finding: Top drifted features: symmetry error=0.173, smoothness error=0.168, fractal dimension error=0.145, worst smoothness=0.141, texture error=0.108

Recommendation: Review moderate drift and evaluate model performance on the shifted slice.

Decision Tree: aggregate risk 18.38 (Low)

Robustness - risk 11.11, status ok

Finding: Baseline accuracy: 0.953.

Finding: Noisy-input accuracy: 0.918.

Finding: Prediction flip rate under perturbation: 0.058.

Recommendation: Keep robustness checks in continuous evaluation as the data distribution changes.

Membership - risk 0.00, status ok

Finding: Confidence-based membership AUC: 0.500.

Finding: Membership advantage proxy: 0.000.

Finding: Train-test mean confidence gap: 0.000.

Recommendation: Continue tracking confidence gaps for each model release.

Calibration - risk 11.70, status ok

Finding: Expected calibration error: 0.047.

Finding: Brier score: 0.047.

Recommendation: Keep calibration in release gates, especially after retraining or data drift.

Pii - risk 30.00, status ok

Finding: PII-like patterns in sampled training text: {'email': 1, 'phone': 1, 'ssn': 0, 'credit_card_like': 0, 'ip_address': 0}.

Finding: PII-like patterns in sampled outputs: {'email': 1, 'phone': 0, 'ssn': 0, 'credit_card_like': 0, 'ip_address': 0}.

Finding: Exact training-output repeats: 0.

Recommendation: Maintain PII scans in data ingestion and output monitoring pipelines.

Unsafe Outputs - risk 33.33, status ok

Finding: Unsafe-output category hits: {'credential_exposure': 1, 'malware_or_intrusion': 0, 'self_harm': 0, 'unsafe_tool_use': 0}.

Finding: Example hits: credential_exposure: Potential secret: api_key=do-not-ship-this-example

Recommendation: Continue red-team prompt evaluation for every prompt/template or model change.

Drift - risk 34.54, status ok

Finding: Maximum feature PSI: 0.173.

Finding: Mean feature PSI: 0.072.

Finding: Top drifted features: symmetry error=0.173, smoothness error=0.168, fractal dimension error=0.145, worst smoothness=0.141, texture error=0.108

Recommendation: Review moderate drift and evaluate model performance on the shifted slice.